Slide Set 7: Causal Inference

Pearl's Ladder · Potential Outcomes · DAGs & Do-Calculus · Instrumental Variables · Causal Discovery

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Outline

1. Pearl's Ladder of Causation
2. Rubin's Potential Outcomes Framework
3. DAGs for Causal Reasoning
4. Unmeasurable Confounders & Instrumental Variables
5. Causal Discovery

Correlation Is Not Causation

Classic confusions:

- Ice cream sales and drowning rates are correlated (confounder: hot weather)
- Shoe size predicts reading ability in children (confounder: age)
- Hospitals have higher mortality (selection: sicker patients go to hospitals)

The key distinction: Statistics asks *what we see*, $P(Y|X)$. Causality asks *what would happen if we intervened*, $P(Y|\mathrm{do}(X = x))$. These are fundamentally different objects.

Why Does This Matter for ML?

Standard supervised learning minimises $\mathbb{E}[\ell(f(X), Y)]$. Fine for prediction under the *same* distribution, but:

- A model trained on observational data learns $P(Y|X)$, not the underlying causal mechanism
- Under **distribution shift** or **intervention**, predictions fail
- **Policy evaluation** requires causal quantities

Example: "patient in ICU" predicts low mortality observationally, but sending patients to the ICU *reduces* mortality causally.

Pearl's Ladder of Causation

Rung 3 — Counterfactual
"What if I had done X instead?"

↑

Rung 2 — Intervention
"What if I do X?"

↑

Rung 1 — Association
"What if I see X?"

Rung 1 — Association: Pure statistics. Query: $P(Y|X = x)$ from observational data.
"Patients taking drug X have better outcomes."

Rung 2 — Intervention: Fix X externally, breaking its causal parents. Query: $P(Y|\mathrm{do}(X = x))$.
"What happens if we prescribe drug X?"
Cannot be read from observational data without additional structural assumptions.

Rung 3 — Counterfactual: Given that patient i took X and died, what would have happened had they *not* taken it? Query: $P(Y_{x'} \mid X = x,\ Y = y)$.
Requires a structural causal model (SCM).

Key Result (Pearl)

No amount of data alone can climb the ladder. Each rung requires additional structural assumptions.

Potential Outcomes Notation

Setup: binary treatment $T_i \in \{0, 1\}$, outcome Y_i, covariates $X_i \in \mathbb{R}^p$.

Potential Outcomes (Rubin, 1974)

For each unit i, define:

- $Y_i(1)$: outcome *if* i receives treatment
- $Y_i(0)$: outcome *if* i does not receive treatment

Individual treatment effect: $\tau_i := Y_i(1) - Y_i(0)$

Fundamental Problem of Causal Inference

We observe only *one* potential outcome per unit:

$$Y_i^{\text{obs}} = T_i\, Y_i(1) + (1 - T_i)\, Y_i(0)$$

$Y_i(1 - T_i)$ is unobserved.

Key estimands:

Quantity	Definition
ATE	$\mathbb{E}[Y(1) - Y(0)]$
ATT	$\mathbb{E}[Y(1) - Y(0) \mid T = 1]$
ATC	$\mathbb{E}[Y(1) - Y(0) \mid T = 0]$
CATE	$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$

SUTVA (Stable Unit Treatment Value Assumption):

1. No interference: $Y_i(T_i)$ does not depend on T_j for $j \neq i$
2. No hidden versions of treatment

Observed data: $\{(X_i, T_i, Y_i^{\text{obs}})\}_{i=1}^n$.
Goal: identify ATE or $\tau(x)$ from this.

Identification: From Observational to Causal

Naive Estimator — Biased

$$\mathbb{E}[Y|T{=}1] - \mathbb{E}[Y|T{=}0] = \mathrm{ATT} + \underbrace{\mathbb{E}[Y(0)|T{=}1] - \mathbb{E}[Y(0)|T{=}0]}_{\text{selection bias}}$$

Selection bias $\neq 0$ when treated and control differ in their potential outcomes.

Ignorability / Unconfoundedness

$$\{Y(0),\ Y(1)\} \perp\!\!\!\perp T \,\Big|\, X$$

Conditional on observed covariates X, treatment is "as good as random."

Identification under ignorability + overlap:

Assume ignorability and **overlap** (positivity): $0 < P(T = 1 \mid X = x) < 1$ for all x.

Then $Y(t) \perp\!\!\!\perp T \mid X$ implies:

$$\mathbb{E}[Y(t) \mid X = x] = \mathbb{E}[Y^{\mathrm{obs}} \mid T = t,\ X = x]$$

ATE via the Adjustment Formula

$$\mathrm{ATE} = \mathbb{E}_X\Big[\mathbb{E}[Y|T{=}1, X] - \mathbb{E}[Y|T{=}0, X]\Big]$$

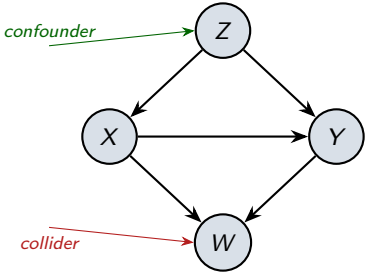
Structural Causal Models and DAGs

Structural Causal Model (SCM)

An SCM $\mathcal{M}$ consists of:

- Endogenous variables $V = \{X_1, \ldots, X_p\}$
- Exogenous noise $U = \{U_1, \ldots, U_p\}$
- Structural equations: $X_i := f_i(\mathrm{Pa}(X_i),\ U_i)$

The *causal graph* G has an edge $X_j \to X_i$ iff $X_j \in \mathrm{Pa}(X_i)$.



$\mathrm{Pa}(Y) = \{X, Z\}, \quad \mathrm{De}(Z) = \{X, Y, W\}$

$P = P(Z)\, P(X|Z)\, P(Y|X, Z)\, P(W|X, Y)$

Markov Factorization:

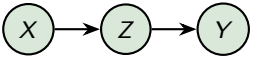
$$P(X_1, \ldots, X_p) = \prod_{i=1}^{p} P(X_i \mid \mathrm{Pa}(X_i))$$

Each variable is independent of its non-descendants given its parents.

The Three Elementary Path Types

Chain (Pipe):

$$X \to Z \to Y$$

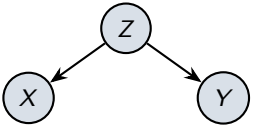


$X \not\perp\!\!\!\perp Y$ (marginally)
$X \perp\!\!\!\perp Y \mid Z$ (Z *blocks*)

Information flows $X \to Z \to Y$. Conditioning on Z *cuts* the flow.

Fork (Common Cause):

$$X \leftarrow Z \to Y$$

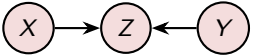


$X \not\perp\!\!\!\perp Y$ (marginally)
$X \perp\!\!\!\perp Y \mid Z$ (Z *blocks*)

Z is a *confounder*. Conditioning on Z removes the spurious association.

Collider:

$$X \to Z \leftarrow Y$$



$X \perp\!\!\!\perp Y$ (marginally)
$X \not\perp\!\!\!\perp Y \mid Z$ (Z *opens!*)

Collider bias: conditioning on a common effect *induces* dependence. E.g. conditioning on "published": talent $\not\perp\!\!\!\perp$ luck.

d-Separation

d-Separation (Pearl, 1988)

A path π between X and Y is *blocked* by Z if:

1. π contains a **chain** $A \to M \to B$ or **fork** $A \leftarrow M \to B$ with $M \in Z$, *or*
2. π contains a **collider** $A \to M \leftarrow B$ with $M \notin Z$ and no descendant of M in Z.

X and Y are **d-separated** by Z (written $X \perp\!\!\!\perp_G Y \mid Z$) if *every* path between them is blocked.

Theorem 1 (Markov + Faithfulness)

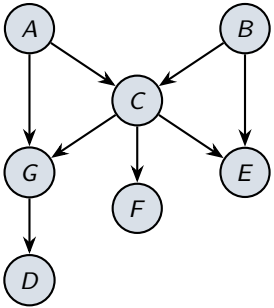
Under the Markov condition:

$$X \perp\!\!\!\perp_G Y \mid Z \implies X \perp\!\!\!\perp_P Y \mid Z$$

Under faithfulness (no cancellations in P):

$$X \perp\!\!\!\perp_P Y \mid Z \iff X \perp\!\!\!\perp_G Y \mid Z$$

Advanced d-Separation & Path Blocking



d-Separation Scenarios

- $A \perp\!\!\!\perp B$
  (Colliders at C and E block all paths between them)
- $A \not\perp\!\!\!\perp B \mid F$
  (Conditioning on F, a **descendant** of collider C, partially opens the path $A \to C \leftarrow B$)
- $A \not\perp\!\!\!\perp D \mid C$
  (C blocks $A \to C \to G \to D$, but the alternative path $A \to G \to D$ remains open)
- $A \perp\!\!\!\perp D \mid G$
  (G fully intercepts all directed paths to D)
- $A \not\perp\!\!\!\perp E \mid C$
  (C blocks the chain $A \to C \to E$, but conditioning on it **opens** the collider path $A \to C \leftarrow B \to E$)
- $A \perp\!\!\!\perp E \mid \{C, B\}$
  (We must also condition on B to block the path that C accidentally opened)

Do-Calculus and Interventional Distributions

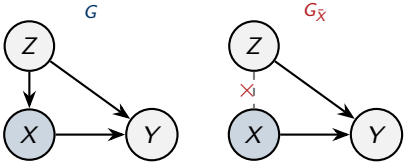
Intervention $\mathrm{do}(X = x)$: remove all incoming edges to X, set $X := x$. Creates the *mutilated graph* $G_{\overline{X}}$.

Truncated Factorization

Under $\mathrm{do}(X = x)$:

$$P(V \setminus \{X\} \mid \mathrm{do}(X{=}x)) = \prod_{V_i \neq X} P(V_i \mid \mathrm{Pa}(V_i))\Big|_{X=x}$$

Mutilated graph example:



Do-Calculus: Three Rules (Pearl)

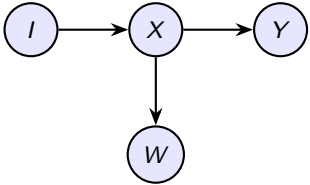
R1 : $P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w)$
if $Y \perp\!\!\!\perp_{G_{\overline{X}}} Z \mid X, W$

R2 : $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w)$
if $Y \perp\!\!\!\perp_{G_{\overline{X}\underline{Z}}} Z \mid X, W$

R3 : $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w)$
if $Y \perp\!\!\!\perp_{G_{\overline{X}\overline{Z(W)}}} Z \mid X, W$

Completeness: any identifiable query can be derived using these three rules.

Applying Do-Calculus: Step-by-Step



Graph Structure

- I: Instrument (e.g., Encouragement)
- X: Treatment (e.g., Drug)
- Y: Outcome (e.g., Recovery)
- W: Side Effect (e.g., Hair loss)

The Derivation

$$P(y \mid \mathrm{do}(x), \mathrm{do}(w), i)$$

$$= P(y \mid \mathrm{do}(x), \mathrm{do}(w))$$

Rule 1: $Y \perp\!\!\!\perp I \mid X, W$ in $G_{\overline{XW}}$

$$= P(y \mid \mathrm{do}(x))$$

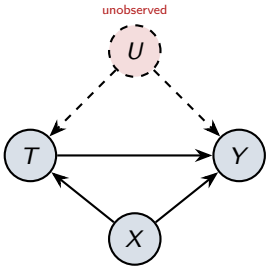
Rule 3: $Y \perp\!\!\!\perp W \mid X$ in $G_{\overline{XW}}$

$$= P(y \mid x)$$

Rule 2: $Y \perp\!\!\!\perp X$ in $G_{\underline{X}}$

When Backdoor Fails: Hidden Confounders

Setup: treatment T, outcome Y, observed X, *unobserved confounder* U.



Backdoor path $T \leftarrow U \rightarrow Y$ exists. U unobserved $\Rightarrow$ cannot block it.

Consequence:

$$\mathbb{E}[Y|T{=}1]{-}\mathbb{E}[Y|T{=}0] = \mathrm{ATE}{+}\text{confounding bias}$$

Controlling for X does *not* remove bias from U.

What can we do?

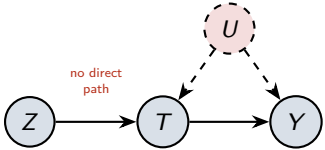
- **Sensitivity analysis:** how large must U's effect be to explain away the result?
- **Partial identification:** compute *bounds* on the ATE (Manski)

Instrumental Variables: Setup and Assumptions

Valid Instrument Z

Z is a valid instrument for the effect of T on Y if:

1. **Relevance:** $Z \not\perp\!\!\!\perp T$ (Z moves treatment; $\mathrm{Cov}(Z, T) \neq 0$)
2. **Exclusion:** $Z \to Y$ only through T (Z has no direct effect on Y)
3. **Independence:** $Z \perp\!\!\!\perp U$ (instrument is unconfounded with outcome)



Wald Estimator (binary $Z \in \{0, 1\}$):

Under the IV assumptions:

$$\hat{\beta}_{IV} = \frac{\mathbb{E}[Y|Z{=}1] - \mathbb{E}[Y|Z{=}0]}{\mathbb{E}[T|Z{=}1] - \mathbb{E}[T|Z{=}0]}$$

Equivalently: $\hat{\beta}_{IV} = \mathrm{Cov}(Y, Z) \,/\, \mathrm{Cov}(T, Z)$.

What does this estimate?
For a linear structural model $Y = \beta T + U$:
$\hat{\beta}_{IV} \xrightarrow{p} \beta$ (the true causal effect).

For *binary* T and the Heterogeneous Treatment Effect Framework, the situation is more subtle: the Wald estimator does *not* recover the ATE in general. It recovers the **LATE**, the effect only for a specific subpopulation called *compliers*.

Compliance Types and the LATE Theorem

Compliance types (binary Z, binary T):

Type	$T(Z{=}0)$	$T(Z{=}1)$	**First stage**
Complier	0	1	Yes
Always-taker	1	1	No
Never-taker	0	0	No
Defier	1	0	Yes (reversed)

Monotonicity (Imbens & Angrist, 1994)

$$T_i(Z{=}1) \geq T_i(Z{=}0) \quad \forall\, i$$

Excludes defiers: the instrument can only push treatment in one direction.

Theorem 2 (LATE Theorem (Angrist, Imbens, Rubin 1996))

Under relevance, exclusion, independence, and monotonicity:

$$\hat{\beta}_{IV} \xrightarrow{p} \text{LATE} = \mathbb{E}\Big[Y(1) - Y(0) \,\Big|\, \textit{complier}\Big]$$

Proof sketch:

$$\begin{aligned}
&\mathbb{E}[Y|Z{=}1] - \mathbb{E}[Y|Z{=}0] \\
&= \mathbb{E}[Y(T(1)) - Y(T(0))] \quad \text{(exclusion)} \\
&= \mathbb{E}[Y(1) - Y(0) \mid \text{complier}] \cdot P(\text{complier})
\end{aligned}$$

Dividing by $\mathbb{E}[T|Z{=}1] - \mathbb{E}[T|Z{=}0]$ gives LATE.

IV Identifies LATE, Not ATE (Manski Bounds)

LATE $\neq$ ATE in general:

- LATE is the effect only for *compliers*
- Cannot identify the effect on always-takers or never-takers from IV alone

Manski (1990) Sharp Bounds

Without assumptions beyond SUTVA, assume the outcome is bounded such that $Y \in [0, M_Y]$. Let $\pi_t = P(T = t)$.

The ATE is sharply bounded in the interval $L \leq \text{ATE} \leq U$:

$$L = \mathbb{E}[Y \mid T{=}1]\pi_1 - \mathbb{E}[Y \mid T{=}0]\pi_0 - \pi_1 M_Y$$
$$U = \mathbb{E}[Y \mid T{=}1]\pi_1 - \mathbb{E}[Y \mid T{=}0]\pi_0 + \pi_0 M_Y$$

Tighter bounds under monotone treatment response ($Y_i(1) \geq Y_i(0)$ for all i): the bounds narrow considerably but remain partial.

Under IV + monotonicity + MTR:

$$\text{ATE} \in [\text{LATE} - \Delta,\ \text{LATE} + \Delta]$$

where Δ depends on the proportion of always-takers and never-takers.

Summary: What IV Does and Does Not Do

- ✓ Removes confounding bias
- ✓ Identifies LATE (compliers only)
- × Does *not* identify ATE without extra assumptions

Markov Equivalence and CPDAGs

Markov Equivalence

Two DAGs G_1 and G_2 are **Markov equivalent** if they entail the same conditional independencies: $\mathcal{I}(G_1) = \mathcal{I}(G_2)$.

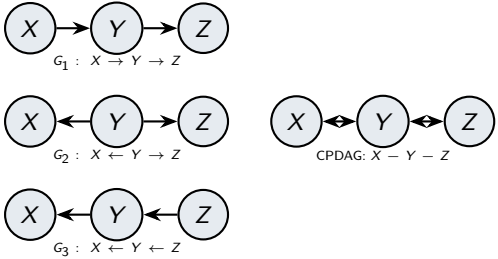
Theorem 3 (Verma & Pearl, 1991)

$G_1 \sim G_2$ iff:

1. *Same **skeleton** (undirected edges)*
2. *Same **v-structures** (unshielded colliders $A \to C \leftarrow B$ with no A-B edge)*

CPDAG (Completed Partially Directed Acyclic Graph): the unique representative of the equivalence class.

Example: three equivalent DAGs for $X - Y - Z$



All three share skeleton X-Y-Z and have no v-structure.

Non-equivalent: $X \to Z \leftarrow Y$ (v-structure at Z).

The Causal Discovery Problem

So far: we assumed the causal graph G is *known*.
Now: can we *learn* G from i.i.d. data $\{X^{(1)}, \ldots, X^{(n)}\}$?

Three fundamental challenges:

1. **Markov equivalence:** observational data identifies at best the CPDAG, not a unique DAG.
2. **Faithfulness:** algorithms require that the observed independence pattern is *exactly* what G entails — violated in near-cancellation regimes.
3. **Finite samples:** CI tests have power < 1, especially for high-order conditioning sets.

Faithfulness Assumption

P is *faithful* to G if every conditional independence in P is entailed by d-separation in G:

$$X \perp\!\!\!\perp_P Y \mid Z \implies X \perp\!\!\!\perp_G Y \mid Z$$

Geometry of violations: Faithfulness fails when path effects *cancel exactly*, e.g. in a chain $X \to Y \to Z$: $\beta_{XY} \cdot \beta_{YZ} = 0$ exactly. The set of parameters where this holds has **Lebesgue measure zero**, generically satisfied.

Practical concern: *Near-faithfulness violations* (small but nonzero path products) cause finite-sample errors. High-dimensional settings are most susceptible.

The PC Algorithm

PC Algorithm (Spirtes, Glymour, Scheines 1993) recovers the CPDAG under Markov + Faithfulness.

Phase 1: Skeleton Recovery

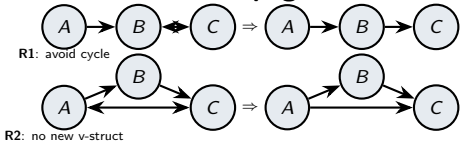
1. Start with the complete undirected graph K_p
2. For each pair (X_i, X_j), test $X_i \perp\!\!\!\perp X_j \mid S$ for $S \subseteq \mathrm{Adj}(X_i)$, $|S| = 0, 1, 2, \ldots$
3. Remove $X_i - X_j$ if a separating set $\hat{S}_{ij}$ is found; record $\hat{S}_{ij}$

Phase 2: V-structure Orientation

For each unshielded triple $X_i - X_k - X_j$ (no X_i-X_j edge):

- If $X_k \notin \hat{S}_{ij}$: orient $X_i \to X_k \leftarrow X_j$ (v-structure)
- Otherwise: leave undirected

Phase 3: Meek's Propagation Rules



Complexity: $\mathcal{O}(p^{|\hat{S}|+2})$ CI tests. Tractable in sparse graphs.

Guarantee: under Markov + Faithfulness, PC returns the true CPDAG.

Causal Discovery: Extensions and Identifiability

Identifiability beyond the CPDAG:
PC outputs the equivalence class, not a unique DAG. Additional structural assumptions allow full identification:

- **LiNGAM** (Shimizu et al. 2006): $Y = \alpha X + \varepsilon$, *non-Gaussian* ε.
  Asymmetry of higher-order moments identifies $X \to Y$ vs $Y \to X$ uniquely.
- **ANM** (Hoyer et al. 2009): $Y = f(X) + \varepsilon$, $\varepsilon \perp\!\!\!\perp X$.
  The reverse model $X = g(Y) + \eta$ typically fails $\eta \perp\!\!\!\perp Y$ — direction is identifiable.
- **NOTEARS** (Zheng et al. 2018): continuous optimisation (DAG constraint via matrix exp). Scales to large p but assumes linearity.

CI tests used in practice:

Setting	Test
Linear Gaussian	Partial correlation (t-test)
Non-parametric	Kernel CI test (KCI, HSIC)
Discrete	χ^2 test, G-test

Algorithm landscape:

Algorithm	Key assumption
PC	Markov + Faithfulness
FCI	PC + allows latent variables
GES	Score-based (greedy BIC)
LiNGAM	Linear + non-Gaussian noise
ANM	Additive noise, general f

Output: PC/FCI/GES $\to$ CPDAG (or PAG); LiNGAM/ANM $\to$ full DAG.